

Shrouk Mohamed

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EDUCATION

Bachelor of Computer Science, Ain Shams University 2022 – 2026 (Expected)
Related Courses: Data Structures, Database Management Systems, Object-Oriented Programming, Artificial Intelligence, Probability and Statistics, Network, Computer Architecture, Operating System.
CGPA: 3.77/4.0

PROJECTS

- FOS Operating System** | *C, Eclipse* | [link](#) Oct 2024 – Dec 2024
- Engineered core memory management functions, including dynamic memory allocation and paging, to optimize RAM utilization.
 - Designed system calls and shared memory areas with locking mechanisms to ensure efficient and secure process interaction.
 - Implemented a Priority Round Robin Scheduler to enhance CPU process management and improve multitasking.
- Fraud Detection** | *Python, Machine Learning, Pandas, Scikit-learn* | [link](#) April 2024 – May 2024
- Developed a machine learning system to classify transactions as fraudulent or legitimate using a historical dataset.
 - Built robust models, including Logistic Regression, Random Forest, and XGBoost, achieving high accuracy in fraud detection.
 - Applied feature engineering and preprocessing techniques to enhance model performance and handle class imbalance.
 - Evaluated models using metrics such as precision, recall, and F1-score to optimize detection of fraudulent activities.
 - Deployed the trained model for real-time monitoring and automated periodic retraining with updated data.
- House Management System** | *Java, File Storage* | [link](#) April 2023 – May 2023
- Developed a standalone Java application for managing house listings, brokers, and customers with a structured, user-friendly system.
 - Implemented a file-based storage system using serialization for efficient data retrieval and management without a database.
 - Integrated role-based authentication to ensure secure access and streamlined interactions between brokers and customers.

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL, JavaScript
Developer Tools: Git, Docker, VS Code, Visual Studio, Eclipse
Libraries: pandas, NumPy, Matplotlib, TensorFlow, PyTorch

ACHIEVEMENTS

- Participated in the **ECPC, Egyptian Collegiate Programming Qualifications Contest 2024**, utilizing C++, data structures, and algorithms. [Certificate](#)

EXTRACURRICULAR ACTIVITIES

- APPLAI** Oct 2023-Present
- Computer Vision Member:** Contributed to research and development in computer vision, working on projects involving image processing and deep learning.
 - Instructor in the Computer Vision Winter Workshop 2024/2025:** Delivered sessions on fundamental and advanced computer vision topics, guiding participants through hands-on implementations and real-world applications.
- RoboTech**
- Best AI Member - Mid-Year Workshop:** Delivered on-stage presentations on Linear Regression and Polynomial Regression algorithms, supported attendees during three sessions on machine learning topics (e.g., Data Cleaning, Decision Tree, KNN, Logistic Regression), worked on five+ datasets, and led a project integrating AI and Electrical concepts.